

Airport runway scheduling

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Abstract Airport runway optimization is an ongoing challenge for air traffic controllers. Since demand for air-transportation is predicted to increase, there is a need to realize additional take-off and landing slots through better runway scheduling. In this paper, we review the techniques and tools of operational research and management science that are used for scheduling aircraft landings and take-offs. The main solution techniques include dynamic programming, branch and bound, heuristics and meta-heuristics.

Keywords Aircraft landing scheduling · Aircraft take-off scheduling · Runway sequencing · Review

Mathematics Subject Classification (2000) 90B35

1 Introduction

Airport runway capacity continues to be a limiting factor in the face of rising demand for more flights. Air traffic demand is expected to more than double in Europe and US, and perhaps triple in some regions, over the next 15 years. Meeting this increased

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demand is challenging given that many airports and routes are already congested. Although investment in infrastructure may be necessary to increase capacity at airports, making the most efficient use of the current infrastructure through improved planning and scheduling is an important part of the solution. According to [Soomer and Franx \(2008\)](#), significant improvements have already been achieved in enhancing en-route traffic capacity, thereby producing a shift in the air traffic bottleneck from en-route segments to airport capacity.

In this paper, we focus on runway scheduling. One of the main factors that determines the runway throughput at an airport is the required minimum separation between aircraft during landing and take-off. The dependency of separation on the leading and trailing aircraft type make the sequencing and scheduling of landings and take-offs an important and non-trivial problem. Careful sequencing and scheduling can reduce the number of long separation times thereby opening up opportunities for new landing or take-off slots.

The Aircraft Landing Problem (ALP) and the Aircraft Take-off Problem (ATP) seek to determine the sequence of aircraft landing on, or taking off from, the available runways at airports in order to optimize given objectives, subject to a variety of operational constraints. While optimally sequencing the landing and taking off of aircraft may increase the runway capacity in theory, it may not always be possible to implement these solutions in practice. For this reason, the challenge lies in putting theory into practice, which involves simultaneously handling safety, efficiency, robustness, competitiveness, and environmental issues. This paper reviews the literature on optimizing the use of airport runways, which has mainly appeared during the past three decades. We also identify some key areas for future research.

The remainder of this paper is organized as follows. Some basic concepts relating to the ALP and ATP are explained in Sect. 2. Section 3 briefly describes general operational constraints, possible objective functions and some modelling techniques for the ALP and ATP. A review of the application of various exact and approximate algorithms for the ALP and ATP are provided in Sects. 4 and 5, respectively. Finally, some conclusions and suggestions for further research are presented in Sect. 6.

2 Basic concepts

An Air Traffic Management (ATM) system aims to assure safety and efficiency of air traffic flows by establishing a set of services. Three types of facilities control the aircraft between the airport used for take-off and the airport used for landing. These are the airport traffic control tower, terminal airspace control centre and en-route control centre.

The *airport traffic control tower* is responsible for ground traffic control, take-off and landing control within about 5 nautical miles (nm) and 3,000 ft above ground level from the airport. The usual responsibilities of controllers in the tower are: clearance delivery, gate hold, ground control, ground planning, and runway control. The *terminal airspace control centre*, which is also called an *approach control airspace* or *terminal radar approach Control*(TRACON), handles departures and arrivals up to 40 nm and 10,000 ft from the airport. The *en-route control airspace*, which is also

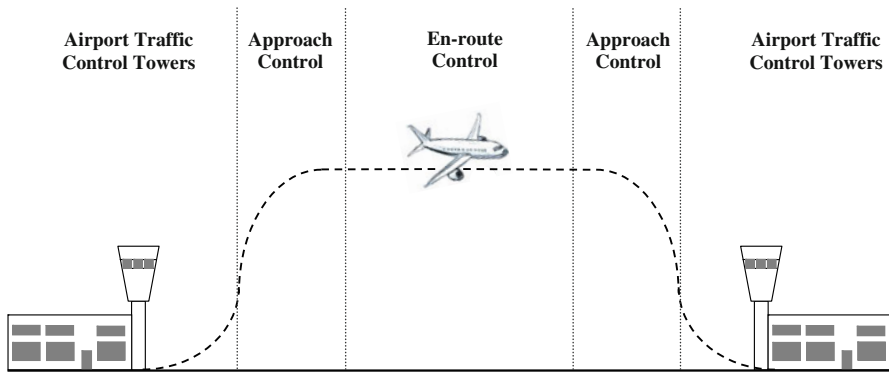


Fig. 1 Air traffic control segments

named the *Air Route Traffic Control Centre* (ARTCC) handles the traffic flow outside the terminal manoeuvring area (see Fig. 1). For further details, we refer to [de Neufville and Odoni \(2003\)](#).

The airspace is divided into a number of geographical regions of varying size, known as sectors. A sector can be defined as a volume of airspace managed by a team of controllers. Generally, sectors which handle high-flying en-route traffic are much larger than the busy sectors which handle a large amount of climbing and descending traffic. Busy airspace, such as in the London area of the UK, can be subdivided into super-low, low, high, and super-high sectors according to altitude. As a flight proceeds through the airspace, responsibility passes from one sector team to the next. There is a limitation on the number of aircraft that can fly in each sector at any given time, which depends on several factors such as safety, flight geometry, controllers' workload, weather, surveillance equipment in use, and the training or experience of the air traffic controllers ([Filar et al. 2001](#)). It appears that dividing the airspace into smaller sectors may help in dealing with the increasing traffic demand. However, this approach requires pilots to change radio frequencies more often and increases the controllers' workload because of greater need for co-ordination between sectors ([de Neufville and Odoni 2003](#); [Duke 2009](#)).

2.1 Decision problems

Generally, ALPs/ATPs consist of the *sequencing*, *scheduling*, and *runway-assignment* decisions. The sequencing process determines the sequence by which aircraft land or take-off from the set of feasible sequences, while the aim of scheduling is to assign a scheduled landing time (SLT)/scheduled take-off time (STT) to each aircraft in the sequence, subject to maintaining operational and safety constraints ([Brinton 1992](#); [Ernst et al. 1999](#)). When more than one runway is available for landing or take-off, each aircraft also has to be assigned to a particular runway.

Scheduling of the landing and take-off of aircraft can be divided into three stages: creating an initial schedule, modifying the schedule, and freezing the schedule ([Mesgarpour et al. 2010](#)). For landing aircraft, the initial schedule is based on a

first come first served (FCFS) order, which is landing order that would result if each aircraft could proceed to the runway and land without consideration of other aircraft. The FCFS order requires updating when new aircraft enter the airport landing planner's radar range (the Extended Terminal Manoeuvring Area, or Extended TMA), about 30–40 min before touch down. The second stage considers new aircraft entering the Extended TMA and adjusts the previous landing sequence to produce an improved schedule. Two to three minutes before landing the schedule is frozen as the aircraft is too close to the runway to make further changes to the landing order or landing time (Neuman and Erzberger 1991). A similar concept applies for the take-off problem. The first stage may start after informing the controllers about the approximate time an aircraft will be ready to leave the gate and start its journey to the runway; this is called pushback. Modifications to the schedule are made in the second stage when precise times for pushback become available to controllers. The freezing stage begins by entering the aircraft into the holding area. This stage varies between airports based on the taxiway, holding area, and runway configuration.

The ALP and ATP models most commonly studied in the literature deal with the *static*(off-line) case, although some consider the *dynamic*(on-line or real-time) case (see Beasley et al. 2004; Moser and Hendtlass 2007; Veidal 2007). In the static version, the model is solved based on a given set of aircraft, where the complete information on these aircraft is assumed to be available and known. It can be argued that ignoring uncertainties leads to a reasonable approximation because information is fairly predictable close to the time of landing or take-off. Nevertheless, after solving the static problem, it is expected that these solutions are revised over time as new aircraft arrive into the system.

2.2 Time windows

The landing time of an aircraft must lie between its earliest and latest possible landing time, which can be dependent on the technical and operational constraints such as fuel limitation, maximum allowed delay, or maximum or minimum airspeed, although other factors such as runway availability, possible manoeuvres, or meeting a connecting flight can also be taken into account. This time window should be treated as a hard constraint. Moreover, there may be a predefined landing time slot in which case controllers aim to assign a *Scheduled Landing Time*(SLT) to each aircraft so that the SLT lies within the aircraft's slot.

If an aircraft uses a congested route or is destined for a hub airport in Europe, then the Central Flow Management Unit (CFMU) of EUROCONTROL in Brussels assigns a *Calculated Time of Take-Off*(CTOT). The CTOT limits the time at which aircraft enter these congested areas with the aim of smoothing the traffic in the airspace and at the airports (Atkin 2008). Controllers aim to assign a *Scheduled Take-Off Time* (STT) to each aircraft from five minutes before to ten minutes after the CTOT. Therefore, the CTOT defines a time window for take-off of the aircraft. However, this is normally treated as a soft constraint because violations are sometimes unavoidable.

In theory, there can be multiple non-overlapping time windows for an aircraft to land or take-off. Hence, an aircraft could be constrained to land in any one of a collection of specified time intervals (Balakrishnan and Chandran 2006).

2.3 First-come-first-served (FCFS)

The simplest way of sequencing aircraft to land on a single runway is through the *first-come-first-served* (FCFS) discipline. It assigns an SLT to each aircraft based upon the order generated by the estimated landing time (ELT) of the aircraft. The landing planner system calculates the ELT based on the planned arrival route, cruise speed (the most economical or preferred speed), and the standard procedure descent profile. Controllers use the ELT as a reference value in computing delays in the terminal area (Neuman and Erzberger 1991). The order of the aircraft queueing at the holding area is the FCFS order, which provides an Estimated Take-off Time (ETT). While the FCFS rule is fair in terms of the ELTs and ETTs and it simplifies the implementation of operational constraints, FCFS does not necessarily match the preferred landing/take-off order since it does not use important information about the problem (Carr et al. 2000). Moreover, it has been established that FCFS rarely provides the best sequence in terms of runway throughput, average aircraft delay or even average passenger delay (Capri and Ignaccolo 2004).

2.4 Runway capacity and assignment

As stated by Idris et al. (1998a,b), the runway provides the main constraint on capacity in an airport system. Blumstein (1959) introduces the first important analytical model for estimating the capacity of an arrival runway. The runway capacity (maximum throughput) can be defined as the maximum hourly rate of aircraft takeoff or landing operations that can reasonably be accommodated by a single or combination of runways. Capacity is generally dependent on the runway occupancy time, mix of aircraft using the runway, availability of taxiways, aircraft type/performance, spacing between parallel runways, intersection point of runways, mode of operation (segregated or mixed), performance of the ATM systems, weather conditions (visibility, wind strength and direction), and noise restrictions (Bazargan et al. 2002; de Neufville and Odoni 2003). The airport capacity model presented by Newell (1979) shows that capacity is greater when runways are operated in mixed-mode. Since increasing the number of runways is often impractical, air traffic controllers aim to use methods and techniques to maximize the throughput from the available runways.

Airports can operate with different numbers and configurations of runways. These can be a single runway, a number of parallel or intersecting runways, or a combination of these. Assigning a runway to the landing/take-off aircraft is a decision made by controllers. The *runway assignment* is typically dependent on the airport configuration, the direction of arriving aircraft (arrival feeder gate), and departure route of the aircraft which is normally specified by the flight plan (Brinton 1992). While an aircraft approaches the runway, adjustments can be made to the flight plan by assigning the aircraft to an alternative runway, which is known as *runway allocation*, in order to balance both the landing/take-off on each runway and the controllers' workload. Airline preferences such as parking gate location, taxi time between the runway and the gate, and controller considerations such as safety, shorter flight times, and lower workloads can lead a controller to assign a new runway to an aircraft (Isaacson et al. 1997).

2.5 Separation

The prime responsibility of the air traffic controllers is the safety of the flights. Standard vertical and horizontal separations that keep aircraft from becoming dangerously close comprise one of the main ATC safety considerations. The usual minimum vertical separation between civilian aircraft operating in controlled airspace is 1,000 ft. The horizontal separations between aircraft vary depending on the position, type and speed of the aircraft, and possibly other considerations.

One reason for setting minimum aircraft separation is to avoid the effect of vortices generated by the aircraft as a consequence of their lift. A Wake Vortex (WV) is potentially hazardous because of the rolling moment it can impose on a following aircraft. Generally, the WV separation between consecutive aircraft depends on the airspeeds, landing/take-off routes, and types of aircraft, and therefore it is sequence dependent. For example, heavier aircraft generate a greater WV and can tolerate more turbulent air. As a result, a light aircraft following a heavy aircraft requires a greater separation than when the following aircraft is also heavy.

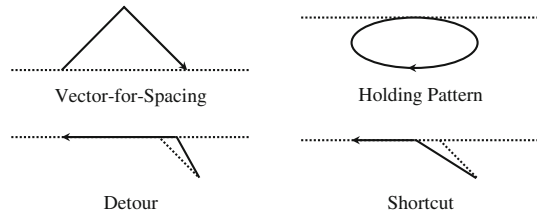
As a consequence of WVs, the International Civil Aviation Organization (ICAO) puts into force separation standards between the leader and the follower aircraft for approach, landing, and take-off to allow safe flight operations. The WV constraints govern the minimum permissible distance between aircraft lined up in sequence on the approach to land on the runway, and the delay before an aircraft can take off from the runway. The separation standard for landing is based on distance, while the take-off separation is based on time. If the scheduling procedure for landing requires a time-based separation instead of distance, the separation distances are typically converted into separation times using a fixed landing speed for the corresponding aircraft type (Beasley et al. 2001). It can be argued that the air speed is not fixed and depends on various factors such as aircraft type, flight level and weather conditions. The separation standard must satisfy the triangle inequality:

$$s_{ik} \leq s_{ij} + s_{jk}, \quad \text{for all aircraft classes } i, j, k, \quad (2.1)$$

where s_{ij} is the WV separation between aircraft classes i and j (Balakrishnan and Chandran 2006).

The simple ICAO's standard international classification of aircraft is based on three weight categories (Heavy, Medium and Light), and using distance separations that are integer (or half integer) numbers of nautical miles thereby making the air traffic controllers' job simpler at the expense of reducing capacity (Tether and Metcalfe 2003). However, in the United Kingdom, the original ICAO three-group scheme has been modified to five groups (Heavy, Upper-Medium, Lower-Medium, Small and Light) to provide more appropriate separations for certain aircraft types. When operating at peak capacity, the WV is often a major concern. It effectively determines runway capacity, and thus limits an airport's capacity in the terminal airspace. The large asymmetries in the minimum required separation can provide an opportunity to reduce airborne delays by shifting aircraft positions in the landing sequence. Further, adjusting the take-off sequence can similarly create additional capacity.

Fig. 2 Holding and manoeuvring techniques



Departure routes also impose separation constraints. Aircraft take-off along specified predefined departure routes called *Standard Instrument Departure* (SID) routes. Aircraft following the same SID route must observe the SID separation. Finally, if two consecutive aircraft belong to different speed groups, then the separation may have to be modified depending on these speed groups.

2.6 Holding and manoeuvres

Controllers may make an aircraft wait (hold) before landing or take-off as a result of traffic congestion, poor visibility, weather conditions, occupancy of the runway, or missed time slots. Holding an aircraft is complicated because of the restriction imposed by a predefined flight plan, congestion, capacity of the holding area, and the dependency of the aircraft's speed on its type, weather conditions, and the altitude.

For departures, aircraft can be held at stands or at specific holding points. While airborne, an aircraft can be held using a number of techniques, namely vector-for-space (VFS), holding pattern (HP), detour, shortcut, or speed control. Figure 2 provides an illustration of the first four techniques. VFS and HP are the main holding procedures controllers use to manage the waiting process in the terminal area (Artiouchine et al. 2008). The VFS manoeuvre consists of a deviation of the aircraft away from its original flight path for a short time so that when it rejoins the flight path the time is later than without the deviation, whereas HPs generate a constant prescribed delay for an aircraft by flying in a loop (see Fig. 2). Several HPs may exist in each terminal area and an aircraft can enter a holding pattern several times. A common HP near airports is known as a *holding stack*, where aircraft are instructed to join a waiting loop at different altitude levels above a feeder fix point (Bianco and Bielli 1993). When an aircraft at the lower level is cleared to leave the hold, the other aircraft are laddered down. Controllers can also use various techniques such as detour (taking a longer route by deviating from the prescribed flight path as illustrated in Fig. 2), shortcut (taking a more direct route, again by deviating from the prescribed flight path as illustrated in Fig. 2), and speed up or slow down (issuing an instruction for the pilot to accelerate or decelerate) to allow aircraft to land before or after their Estimated Landing Times (ELTs).

2.7 Position shifting

In practice, delaying or advancing an aircraft by a large number of places in the FCFS take off/landing sequence is undesirable because of the operating environment.

Dear (1976) introduces *constrained position shifting* (CPS) for the ALP to limit the extent of deviations from the scheduling sequence. CPS defines the maximum number of positions an aircraft can shift in the landing sequence relative to the FCFS order. Specifically, *maximum position shifting* (MPS) uses an integer k to define the maximum deviation of the landing position of an aircraft in the selected landing sequence from its landing position in FCFS; this is also referred to as k -MPS. When k is small, an element of fairness among the aircraft is maintained by reducing the deviation from the FCFS sequence. A small value of k has the added advantage when solving the ALP or ATP that the size of the search space is reduced. As shown by de Neufville and Odoni (2003), the most undesirable landing sequences, such as those with a heavy aircraft followed by a light aircraft with an associated high separation, can be avoided, and delays can be significantly reduced, even with $k = 2$ or $k = 3$. Mesgarpour et al. (2010) consider *constrained time shifting* (CTS), which limits the deviation of landing time of an aircraft from that obtained using the FCFS sequence, where the limit can be dependent on aircraft type.

Re-sequencing becomes increasingly difficult as aircraft become closer to landing. *Relative position shifting* (RPS) takes this into account by defining the maximum number of position shifts (either backward or forward) of any aircraft in the sequence relative to the position that it occupies in FCFS. The maximum number of positions in RPS can be defined by the air traffic controllers for any subsequence of the landing/take-off sequence. Generally, this value decreases for subsequences near the beginning of the sequence and can be set to zero in the freezing stage (Bianco et al. 1997).

3 Problem definition

This section describes the possible constraints and objective functions for the ALP and ATP that have been considered within the literature. Some approaches for modelling the ALP and ATP are also discussed.

3.1 Assumptions and constraints

The various studies in the literature on airport runway scheduling detail a variety of operational constraints. The previous section discusses separation time, maximum delay, time windows and maximum position shift from the FCFS order. In addition, some researchers sometimes consider precedence constraints, and air-route constraints. Balakrishnan and Chandran (2006) suggest that one aircraft may be given precedence over others due to overtaking constraints, airline preferences, or high priority flights. Tang et al. (2008) describe a scenario where two aircraft flying on the same air-route have to be scheduled in FCFS order to reduce the risk of aircraft collision.

Airports with multiple runways may operate the runways in either segregated-mode or mixed-mode. In segregated-mode, the runway is solely used for either landing or take-off of the aircraft, while mixed-mode allows both landing and take-off on the same runway. Mixed-mode is usually more efficient than segregated-mode since alternating landings and take-offs on the runway is effective in reducing or eliminating delays due to wake vortex constraints (Atkin et al. 2008).

3.2 Objectives

Air transportation has a number of different stakeholders, including ATC, airlines, airports, and government, who each have their own explicit or implicit objectives. As a result, the formulation of the ALP and ATP involves the simultaneous optimization of a variety of objectives that may conflict, which is likely to lead the decision-maker into considering possible tradeoffs. The main (single) objectives are as follows (Idris et al. 1998a; Fahle et al. 2003; Lee and Balakrishnan 2008).

- (a) ATC aims to ensure safety and efficiency of the aircraft. The following are desirable from an ATC perspective:
 - maximizing the runway throughput
 - minimizing the approach time of aircraft before landing
 - minimizing air traffic controllers' workload
 - maximizing fairness among the aircraft
 - minimizing the aircraft taxi-in/taxi-out time
 - minimizing the arrival/departure delay
 - minimizing deviations from an appropriate balance between arrivals and departures.
- (b) The airline's main objectives are:
 - minimizing operating costs (especially fuel costs)
 - minimizing engine run times before take-off
 - maximizing punctuality with respect to landing/take-off time in published timetables
 - minimizing total passenger delays
 - maximizing adherence to airline priorities within their own flights
 - maximizing the connectivity between incoming and outgoing flights.
- (c) The airport priorities are:
 - maximizing punctuality relative to the operating schedule
 - minimizing the need for gate changes due to delays.
- (d) The government preference is:
 - minimizing environmental effects (noise and air pollution).

Typical scheduling objectives used in the literature, similar to those used in production scheduling, are minimizing the average delay (average tardiness), average landing/take-off time (equivalent to average flow time or average completion time), and landing/take-off time of the last aircraft in the sequence (makespan). Delay is usually defined as the deviation of actual landing/take-off time from the estimated landing/take-off time calculated by the FCFS principle, and not based on the aircraft schedule. Delay is a service-based objective, whereas flow time, completion time and makespan are throughput-based objectives.

3.3 Modelling techniques

The literature provides a range of approaches to modelling and solving the ALP and ATP. We identify the core modelling approaches below, and in the following sections we provide more details of some specific algorithms from the literature.

As shown by [Beasley et al. \(2000\)](#) and [Mesgarpour et al. \(2010\)](#), the ALP and ATP can each be formulated as a mixed integer program (MIP). However, both problems are NP-hard and computation time to find an exact solution is likely to grow exponentially with the number of aircraft. As a result, solving the MIP is unattractive for real-time implementation on practical-sized instances.

[Brentnall \(2006\)](#) points out the relationship between the ALP and a machine scheduling problem with sequence-dependent setup times. The objective function can include makespan, and total earliness and tardiness by penalizing early and late jobs in terms of time windows or due dates. Regarding the specifics of the machine scheduling problem, each job corresponds to a landing operation of the aircraft; each machine with capacity one represents a runway; the ready time (release date) of the job corresponds to the estimated landing time (ELT) of the aircraft; the starting time of the job represents the actual landing time (ALT) of the aircraft; the completion time of the job corresponds to the time the aircraft frees the runway; and the sequence-dependent processing time between jobs represents the required separation between aircraft.

The travelling salesman problem (TSP) and the ALP also have resemblances ([Luenberger 1998](#)). The TSP is one of the most intensively studied problems in computational mathematics. The classic TSP is to find a shortest tour that visits every city exactly once, starting and finishing at the same origin city. The single-runway ALP closely resembles an open TSP with time windows, where the tour does not return to the origin. Each city corresponds to an aircraft, intercity distances represent the separations between aircraft, and the time windows for visiting each city are the landing time windows. The multiple-runway problem similarly resembles a multiple-TSP.

Finally, it is natural to consider the ALP/ATP as a queueing system ([Pujet et al. 1999](#); [Idris 2001](#); [Bauerle et al. 2007](#)). Different classes of aircraft correspond to different customer types and the runways are servers. The service time of each customer (aircraft) is the separation time between the aircraft and its successor. Different queueing models can be used to represent the ALP/ATP problem depending on the number of runways available, the mode of operation at each runway (segregated or mixed), and the method of runway allocation by the controllers.

4 Review of aircraft landing problem

In this section, we review the literature on the ALP. The subsections are organized according to the main methodology used in the study. Unless stated otherwise, the problem considered is to schedule the landing of n aircraft each belonging to one of C classes, with separation times defined by the classes of the leader and follower aircraft.

4.1 Dynamic programming

Dynamic Programming (DP) is a general optimization technique for making sequential decisions. Almost all ALPs can be usefully modelled as DP problems because the algorithms can evaluate current partial solutions independently of the exact sequencing decisions used to form these solutions. Beginning with the early work

of Psaraftis (1978), there have been several attempts to develop efficient dynamic programming algorithm for the ALP.

Psaraftis (1978, 1980) considers a simplified version of the ALP in which all aircraft are available to land immediately. As an objective function, he considers throughput as measured by the landing time of the last aircraft, and delay as measured by the sum of landing times of the aircraft. He develops backward dynamic programming algorithms that have as state variables the number of aircraft from each class that has not yet been scheduled to land and the class of the last aircraft to land. These algorithms essentially merge the lists of aircraft within the different classes. For the case of a single runway, the DP algorithm has a time complexity of $O(Cn^C)$, where C is the number of classes and n is the number of aircraft. For fixed C , this represents a polynomial time algorithm. Further, the DP algorithm can be adapted to handle CPS without increasing the time complexity. An extension of the algorithm to the case of two runways is also proposed.

Brentnall (2006) develops the dynamic programming approach of Psaraftis in two directions. First, he assumes that aircraft have earliest landing times. For some objective functions including the landing time of the last aircraft and the sum of landing times, he establishes that, within each class, aircraft should be sequenced in non-decreasing order of earliest landing time. Using these properties, he develops two forward DP algorithms: one minimizes the landing time of the last aircraft in $O(Cn^C)$ time, and the other minimizes the sum of landing times in $O(n^{C^2+C+1})$ time. For the second direction, he assumes that all aircraft are circling in several stacks, and the aircraft to land next is to be chosen as one at the bottom of one of these stacks. Using as state variables the number of aircraft from each stack that have been scheduled to land and the stack previously containing the last landed aircraft, forward DP algorithms requiring $O(n^{C^2+K})$ time are proposed for general objective functions, where K is the number of stacks. Brentnall and Cheng (2009) suggest four delay sharing strategies: all delay in hold, delay as late as possible, delay as early as possible, and delay evenly throughout the route. By linking the algorithms and methods to a discrete-event simulation and using several statistical methods, the output from simulations based on Stockholm Arlanda airport is analyzed.

Bayen et al. (2004) propose a model that takes account of the time taken to complete a circuit in a holding stack. They assume that all aircraft belong to a single class. They develop a 5-approximation algorithm for the problem of minimizing the sum of landing time, and a 3-approximation algorithm for minimizing the landing time of the last aircraft. Their algorithms combine dynamic programming and the rounded solution of a linear program.

An alternative dynamic programming approach is introduced by Balakrishnan and Chandran (2006). They consider the problem of minimizing the landing time of the last aircraft, and impose CPS, precedence constraints between aircraft and arrival time-window constraints. The problem is formulated as a modified shortest path problem in a network with $O(n(2k+1)^{2k+2})$ arcs, where k is the maximum position shift (see Fig. 3). In addition to a source node s and a terminal node t , the network consists of n stages, where each stage represents an aircraft position in the final sequence. A node at stage σ of the network corresponds to a subsequence of the aircraft of

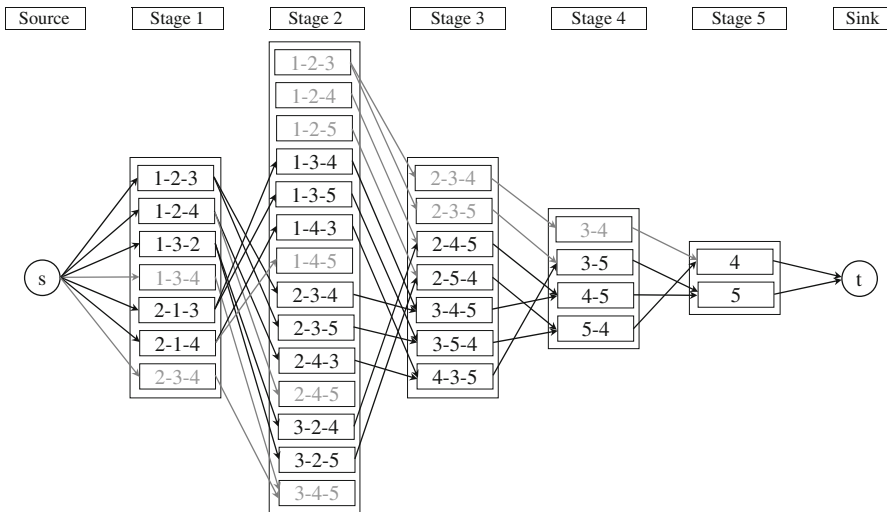


Fig. 3 Pruned network for $n = 5$ and $k = 1$

length $\min\{2k + 1, n - \sigma + 1\}$. If a node at stage σ can be followed by a node at stage $\sigma + 1$, they are connected by a directed arc. The network shown in Fig. 3 for $n = 5$ and $k = 1$ represents all the sequence combinations of possible aircraft assignments to each position. A pruned network, which is significantly smaller than the original network, can be produced by removing nodes which are not part of a path from source to sink (shown in grey) or which violate the precedence constraints.

Some notation is needed in our statement of the dynamic programming algorithm of Balakrishnan and Chandran (2006). Let $P(j)$ represent the set of predecessor nodes of each node j . Also, the earliest and latest landing times of the first aircraft in the sequence associated with node j are denoted by $e(j)$ and $l(j)$, respectively. The shortest path length in the network from node s to each node j of the network, which represents the landing time assigned to the first aircraft in the subsequence associated with node j , can be computed using the following forward dynamic programming recursion:

$$T(j) = \max \left\{ e(j), \min_{i \in P(j): T(i) \leq l(i)} \{T(i) + s_{ij}\} \right\}.$$

Note that the density of the pruned network is significantly smaller than the worst-case complexity expression in practical instances. Also, since the basic network remains the same for any given n and k , it can be stored and retrieved when required. The computational experience of Balakrishnan and Chandran (2006) is based on an implementation of their algorithm on realistic data from the arrival flow at Denver International Airport. The algorithm exhibits small computation times for instances with up to 50 aircraft and with $k \leq 3$.

Building on the approach described above, Chandran and Balakrishnan (2007) introduce a dynamic programming algorithm to compute the tradeoff curve between

the robustness and throughput. Robustness is interpreted as the probability of not violating any separation constraints. Their proposed algorithm is computationally efficient with a time complexity of $O(n(L/\epsilon)^3)$, where L is the largest difference between the latest and the earliest landing time over all aircraft, and ϵ is the interval used when discretizing time.

More recently, [Lee and Balakrishnan \(2008\)](#) extend the previous framework proposed by [Balakrishnan and Chandran \(2006\)](#) and [Chandran and Balakrishnan \(2007\)](#) by presenting a dynamic programming algorithm for minimizing the sum of landing costs of an arrival schedule. They use this approach firstly for minimizing total delay, which is equivalent to minimizing the sum of landing times, and secondly for minimizing fuel cost, where the strategy of speeding up some aircraft at the expense of burning extra fuel is explored. The study shows that speeding up to allow a landing up to 3 min earlier than normal is often advisable. By generating 1,000 problem instances of 30-aircraft sequences using a Poisson distribution, the tradeoff between minimizing the average delay and maximizing the throughput as objectives, which are not necessarily aligned, is investigated. Results shows that the significant improvements in the average delay are achievable through decreasing the throughput so that the delay becomes relatively small.

4.2 Branch-and-bound

[Brinton \(1992\)](#) introduces one of the first branch-and-bound approaches for the ALP and the runway assignment problem. Static, dynamic, and depth limiting methods are used to reduce the number of tree branches that need to be searched. The objective function is a weighted combination of various costs, although the proposed implicit enumeration algorithm does not depend on which costs are included. The methodology is the foundation of the Traffic Management Advisor (TMA) tool of the Center TRACON Automation System (CTAS) developed at NASA Ames Research Center. The implementation of runway and sequence optimization is discussed, although there are no detailed computational results.

[Abela et al. \(1993\)](#) propose another branch-and-bound algorithm based upon a 0–1 mixed integer programming formulation for the single-runway ALP. The objective function has a cost for each aircraft that is attributed to either speeding up or holding. Furthermore, they propose a genetic algorithm in which two heuristic operators, force feasible and squash, are used to ensure that the schedule after crossover satisfies the minimum separation times. Computational results for problem instances with up to 20 aircraft are presented.

[Ernst et al. \(1999\)](#) design a branch-and-bound algorithm and a local search heuristic for the ALP with single and multiple runways, where the objective function comprises penalty costs for landing before and after target times. A key component of their approach is a specialized simplex algorithm for determining a landing schedule, given on a partial order of the aircraft. They develop a heuristic-based problem space search which comprises their simplex algorithm, a constructive based heuristic to generate a good sequence, and a genetic algorithm to search the perturbation space. The heuristic algorithm and simplex method are used to obtain upper and lower bounds

for the branch-and-bound algorithm. Various devices such as tightening intervals, upper-bound-based fixing, and fixing based on data are used as pre-processing methods to improve the performance of the branch-and-bound algorithm. An extended version of the algorithm can be used for multiple runways. The OR-Library data sets (Beasley 1990) are used to evaluate the heuristic and exact algorithms on instances involving up to 50 aircraft on both a single runway and multiple runways.

In addition to providing an extensive literature overview on the ALP, Beasley et al. (2000) design branch-and-bound algorithms by employing linear programming (LP)-based tree search approaches for both single- and multiple-runway problems. They use the same objective function of Ernst et al. (1999) in which there are penalties for landing before and after target times. Their formulation of the problem is based on that introduced earlier by Abela et al. (1993). However, some additional constraints are proposed in order to reduce the zero-one space of the mixed integer formulation and strengthen the LP relaxation. The ALP is solved optimally for the problem instance found in the OR-Library Beasley (1990) involving up to 50 aircraft and four runways.

4.3 Branch-and-price

The multiple-runways version of the ALP is addressed by Wen (2005) and Wen et al. (2005) using a column generation approach. They also adopt the objective function of Ernst et al. (1999) in which there are penalties for landing before and after target times. The ALP is formulated as a set partitioning problem with side constraints. They develop and test a branch-and-price exact algorithm using the problem instances of Beasley (1990), which include 50 aircraft and four runways. Their computational results show that the linear relaxation of the set partitioning model provides a better lower bound than the linear relaxation of the mixed integer program. Further, the branch-and-price approach solves all instances while generating less than 450 columns and exploring no more than 12 nodes in the search tree.

4.4 Heuristics

Based on the observation that constrained position shifting can significantly reduce the number of candidate landing sequences, Dear and Sherif (1989, 1991) present an enumerative heuristic for the static and dynamic ALP. Computational results involving up to 500 aircraft and one runway show the smaller delays obtained under the heuristic than for a FCFS approach.

Bianco et al. (1999) suggest the ALP as an application of the single machine scheduling problem with release dates and sequence dependent processing times to minimize the sum of completion times (often referred to as problem $1|r_j seq-dep|\sum C_j$). They develop a dynamic programming formulation that is of theoretical interest because one of the state variables is the current set of scheduled jobs, and three lower bounds. Further, they propose two heuristic algorithms. The first is an $O(n^2 \log n)$ construction procedure that builds a schedule by adding jobs to the current partial sequence, and the second is based on an $O(n^4)$ insertion approach. The effectiveness of the proposed heuristics is evaluated using randomly generated test instances, and two realistic ALP problem instances that include 30 and 44 commercial aircraft belonging to four weight

classes. The schedules span a period lasting about 40 min, and are evaluated according to total aircraft landing time, and maximum and average landing delay. Because they do not consider CPS in their model, some aircraft are subject to excessive delay.

4.5 Genetic algorithms

Genetic algorithms (GAs) provide a popular approach for tackling sequencing problems, with many successful applications reported in the literature. Hence, applying a GA to the ALP is worthy of investigation. Generally, a chromosome represents the order of the aircraft in a landing sequence. As an example of an alternative encoding, [Beasley et al. \(2001\)](#) use the assigned landing times of the aircraft.

[Stevens \(1995\)](#) provides one of the first and the simplest application of GA for minimizing the total penalty for landing before and after specified target times. Operational constraints include a constant three-minute separation time and an earliest landing time of three minutes before the target landing time. Results of computational tests are presented for ten problem instances involving two runways and up to 40 aircraft over a one-hour scheduling horizon.

Based on the GA of [Stevens \(1995\)](#), [Ciesielski and Scerri \(1997, 1998\)](#) compare two GA implementations for the dynamic/on-line ALP in terms of the percentage of valid solutions obtained and the best objective function value. The first algorithm builds the new schedule from scratch, and the second seeds it from the population left at the end of the last problem by removing landed aircraft and inserting new aircraft into the scheduling horizon. Their computational results are presented for two data sets involving 28 aircraft in a 37 min period and 29 aircraft in a 38 minute period on two runways.

[Cheng et al. \(1999\)](#) design four different genetic schemes for the multiple-runway ALP. The first GA uses two chromosomes to encode the landing sequence and the runway assignment, respectively. In the second and third schemes, each chromosome forms a component of priority list for the flights. A fourth approach is based on genetic programming (GP) with chromosomes defined as mathematical operations and functions. They evaluate the four approaches using one instance involving 12 aircraft and three runways. [Hansen \(2004\)](#) builds on the work of [Cheng et al. \(1999\)](#) by examining the efficiency and effectiveness of GA and GP methods in solving ALP. By applying four genetic search methods on four different test scenarios, he shows that the GP method provides the best solutions and these solutions can support controllers in real-time situations.

[Beasley et al. \(2001\)](#) develop a population heuristic for the static single-runway ALP with time-window restrictions. The algorithm aims to minimize the squared deviations from target landing times. Their computational results are for a single problem instance with five classes of aircraft, where the data for the instance are obtained from observations during a busy period at London Heathrow airport.

In a GA implementation of [Capri and Ignaccolo \(2004\)](#) for the static and dynamic ALP, three different objective function formulations are investigated for minimizing delay. Implementations to solve the static ALP consist of the GA, the GA with maximum landing time constraints, the cheapest insertion heuristic of [Bianco et al. \(1997\)](#),

and FCFS. Results from four test problem cases proposed by Bianco et al. (1997) are used to compare the approaches. The performance of the dynamic model is evaluated using four test instances with up to 30 aircraft.

Pinol and Beasley (2006) implement two different population heuristics, scatter search and a bionomic algorithm, for the multiple-runway ALP. Solutions are represented by specifying the position of the landing time within a given time window and a runway assignment for each aircraft. Both a linear and a non-linear objective function are considered. The linear objective has penalties for landings both before and after a target landing time, while the non-linear objective has a positive quadratic penalty for landings after the target time and a negative quadratic penalty for landings before the target time. An infeasibility penalty for violations of the separation constraints between aircraft is considered separately. Computational tests with OR-Library data sets Beasley (1990) indicate that the relative effectiveness of the scatter search and bionomic algorithms depends on whether the linear or non-linear objective function is assumed.

Yu et al. (2009) use cellular automation (CA) to generate a landing sequence for the single-runway ALP. Improvements are made to the landing sequence using a GA with a relaxation operator. The instances of Beasley et al. (2000) are used to test the method. Bencheikh et al. (2009) propose a hybrid method to solve the ALP where ant colony optimization generates the initial population of feasible solutions for the GA. The ALP is formulated as a job shop scheduling problem with partial orders and alternative sequences through “and/or” graphs.

The dynamic ALP is studied by Hu and Chen (2005a,b), and we review their contributions below. In later work, Hu and Di Paolo (2008) introduce a new type of chromosome which defines a 0–1 value matrix based on neighbouring relationships between each pair of aircraft. The binary representation makes it easier to perform an efficient uniform crossover operator. The proposed GA based on a binary representation is compared with the GA with a permutation representation, as introduced by Hu and Chen (2005a) for the static and dynamic versions of the ALP. These authors also design a GA with uniform crossover for the multi-runway ALP using the successor relationship between aircraft to construct the chromosomes rather than the order of the aircraft in the queues Hu and Di Paolo (2009). They compare their method with the GP approach of Hansen (2004) and an extended version of the GA of Hu and Chen (2005a).

Hu and Chen (2005a,b) study GAs for the dynamic ALP using an approach based on receding horizon control (RHC). Figure 4 presents the comparison of the RHC with other optimization strategies (Hu and Chen 2005b). The *off-line* strategy optimizes the static ALP for the entire time horizon. The *one-step-ahead adjustment* modifies the landing sequence for the current time interval given the static solution and current information. The *conventional dynamic optimization* and the *RHC* optimizes ALP over the horizon from the current time to the end of the time horizon or M time intervals ahead respectively, repeating the procedure at the beginning of each interval based on new information. For the GA of Hu and Chen (2005a), the aim is to minimize the airborne delay, which is the deviation of the actual landing time from the earliest landing time. The performance of their GA is compared with the approach of Bianco et al. (1997) and a GA that is based on conventional dynamic optimization (CDO) using test problem instances from Bianco et al. (1997). The performance of the GA

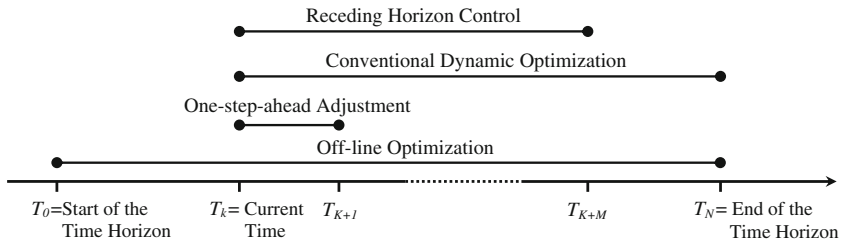


Fig. 4 Different optimization strategies

based on RHC is evaluated further by [Hu and Chen \(2005b\)](#) under different levels of uncertainty and congestion.

4.6 Ant colony optimization

In addition to the use of ant colony optimization (ACO) for generating initial solutions by [Bencheikh et al. \(2009\)](#), [Randall \(2002\)](#) uses ACO alone to tackle the ALP. The objective is to minimize the total penalty associated with aircraft landing before and after specified target times as defined by [Beasley et al. \(2000\)](#). In addition, there are time-window constraints. Six problem instances from [Beasley \(1990\)](#) are used for computational tests. Results show that the quality of the solutions is not as good as those obtained by [Beasley et al. \(2000\)](#).

4.7 Queueing theory

[Bauerle et al. \(2007\)](#) model the ALP as a special queueing system with the incoming aircraft corresponding to customers of different types and separation times between aircraft corresponding to customer service times. The single-runway problem is modelled as an M/SM/1 queueing system, with semi-Markov service times. In addition, they derive the stability condition and the average waiting time. Several routing heuristic strategies are studied and compared with respect to the average delay for assigning aircraft to two runways.

4.8 Comparative studies

Some research studies try to compare different algorithms for solving the ALP. [Fahle et al. \(2003\)](#) consider the simplest model of single-runway ALP with time-window and separation constraints in a static environment. They compare four exact methods. The first two are proposed by [Beasley et al. \(2000\)](#): a mixed integer programming model that uses continuous time, and an integer programming formulation that is modelled using discrete time. The other two methods consist of a constraint programming model, and a satisfiability problem formulation. In addition to these exact methods, they implement two local search heuristics based on descent and simulated annealing. These six methods are evaluated using three problem instances with up to 123 aircraft.

In another study, [Beasley et al. \(2004\)](#) define a displacement problem, which is used in the solution of the dynamic ALP. The rationale of this approach is that unfavourable deviations from the previous solution are penalized in the displacement problem when new information becomes available and the solution is updated. Possible solution approaches for the displacement problem are the LP-based tree search ([Beasley et al. 2000](#)), a heuristic algorithm ([Beasley et al. 2000](#)), and a population heuristic ([Beasley et al. 2001](#)). Their computational results are presented for two sets of test problems involving up to 500 aircraft and five runways.

5 Review of aircraft take-off problem

The ALP has attracted much greater research interest than the ATP for which studies are quite scarce. [Idris et al. \(1998a,b\)](#) study the flow constraints and the dynamics of airport systems. Specifically, they analyze the departure flow at Logan airport as a complex queuing system. As aircraft compete for limited resources such as gates, ramp, taxiways and runways, queues are created in various parts of the airport. They conclude that the runway is the main constraint. [Pujet et al. \(1999\)](#) develop an alternative queuing model of the departure system. Their model is evaluated using the runway configuration and traffic data. The intention is to relieve the departure traffic congestion on the ground.

5.1 Dynamic programming

[Craig et al. \(2001\)](#) propose a dynamic programming algorithm for sequencing the take-off of aircraft at one of the simplified holding points at London Heathrow airport. Some possible strategies for sequencing aircraft at the stands are also considered in their research. Based on their model for the ALP [Balakrishnan and Chandran \(2006\)](#), [Balakrishnan and Chandran \(2007\)](#) introduce a dynamic programming algorithm for the ATP. Their approach is also extended for multiple runways and active runway crossing.

5.2 Heuristics

[Anagnostakis and Clarke \(2002, 2003\)](#) investigate a two-stage heuristic algorithm for solving a runway operation planning problem. The first stage aims to maximize the throughput by generating candidate sequences of classes of aircraft, while ignoring the operational constraints. The second stage uses an integer programming model to assign aircraft to class slots in one of the sequences generated in the first phase taking into account the relevant constraints (see also [Anagnostakis 2004](#)).

5.3 Metaheuristics

[Atkin et al. \(2004\)](#) consider an initial model of the simplified ATP at London Heathrow airport including the holding point structure. Based on the results of comparing

different search heuristics, their conclusion is that tabu search performs better than simulated annealing and descent algorithms. A later and more detailed study of [Atkin et al. \(2006\)](#) investigates the effects of the different constraints on scheduling. A key aspect of their algorithms involves the checking of candidate take-off sequences for feasibility, taking into account the current positions of aircraft at the holding points.

[Atkin et al. \(2007\)](#) propose a hybrid approach that uses different search methodologies and a heuristic method to solve the static version of the ATP. Their objective function comprises a weighted sum of delay, a reordering cost (for an aircraft that is moved later in the take-off sequence than in FCFS, the cost is proportional to the number of positions moved), a non-linear cost for violation of a CTOT time window or for scheduling a take-off near to a boundary of the window, and an additional penalty cost for schedules that introduce an excessive delay an aircraft. The model is evaluated using six sets of data from London Heathrow airport. Result shows that the availability of more information about the aircraft taxiing can reduce delays and improve CTOT compliance. In a further study, [Atkin et al. \(2008\)](#) provide some further enhancements to their previous work (see also [Atkin 2008](#)).

5.4 Constraint satisfaction

Using a constraint satisfaction approach for the ATP and the ILOG solver, [van Leeuwen et al. \(2002\)](#) map flights onto activities, and model the taxiways, runways, and exit points of an airport as resources. Also, different type of constraints such as take-off order, time-slot, and separation are listed as temporal or resource constraints in the ILOG environment. Results of applying the model to real data from Prague airport for up to 12 aircraft in a 50- min time interval are also presented. As the problem size gets larger, the model fails to find a solution in reasonable time. [van Leeuwen and van Hanxleden Houwert \(2003\)](#) introduce constraint relaxation techniques to overcome the highly complex or conflicting requirements that have to be considered in practice. The constraints are divided into different sets of soft and hard constraints according to whether they are candidates for relaxation. Additional controller-imposed constraints, time-slot constraints, and runway preferences are considered as soft constraints.

5.5 Combined aircraft landing and take-off problem

[Trivizas \(1998\)](#) introduces a dynamic programming approach for solving optimally the static runway scheduling problem for landings and take-offs based on the CPS concept. The mixed-mode, segregated-mode, and multiple-runway environments are considered. His computational results obtained with actual traffic data and a real airport configuration show that even a modest value such as a maximum position shift of three can increase the runway capacity up to 20% compared to FCFS sequencing.

[Bianco et al. \(2006\)](#) introduce static and dynamic models for scheduling the landing and take-off of aircraft in the terminal manoeuvring area (TMA). The proposed deterministic job shop scheduling model can represent several operational constraints and different runway configurations. The model considers the runway, TMA, inbound and outbound flight paths, holding stacks for landing and holding points for take-off.

The solution method is based on a fast descent heuristic. Experimental results using real data of Milan Malpensa and Rome Fiumicino airports shows that the average delay can be reduced by more than 40% and the TMA capacity may increase up to 30% in comparison to FCFS sequencing.

6 Concluding remarks

Predictions for increasing air traffic over the next 15 years puts pressure on air navigation service providers around the world to improve safety levels, reduce delays, and cut the costs. This is the motivation behind the SESAR (Single European Sky ATM Research) and NextGen (Next Generation Air Transportation System) programs. SESAR is a European air traffic control infrastructure modernization program that aims to eliminate the fragmented approach to European air traffic management, to transform its system, to synchronize all stakeholders, and to federate resources (EUROCONTROL (2009) SESAR Web site). NextGen is the transformation of the entire air transportation system through the use of twenty first century technology to support the current and future demand for aviation services in the United States (FAA, 2009, NextGen Web site).

After reviewing the research studies on ALP and ATP and observing air traffic controllers in a working environment, we would like to highlight some of our findings.

Practical vs theoretical models: As explained above, many theoretical studies may show an increase utilization of the runway capacity, but it may not be possible to implement the models in practice. Often some critical operational constraints in the modelling are ignored, some of the hard constraints in obtaining a solution are relaxed, or required computational resources are unreasonable.

Quick and good versus slow and optimal solution methods: In real situations, controllers can only use algorithms which can quickly (in a matter of seconds) find a good solution (near-optimal). Optimal solutions arising from lengthy computation times are of little practical use.

Defining the objective functions and constraints: Choosing an appropriate objective function for the ALP/ATP is controversial and stakeholders (air traffic control, airports, airlines, and government) may have conflicting criteria. Thus, selecting one or more objective that can satisfy the interests of all parties, or provide an acceptable compromise, is an important first step towards the model being implemented.

Robustness and flexibility: There are different levels of uncertainty associated with the information considered within an ATP/ALP, especially in a dynamic environment. The uncertainty can be caused by weather such as winds and snow, the precision of equipment, as well as the uncertainty in pushback times and taxi times for departing aircraft. However, most studies consider a static rather than a more realistic dynamic environment.

Increasing the number of separation categories: Currently, ICAO classifies aircraft into three categories. Since wake vortex separation is a primary constraint on runway throughput, refining the classification into more classes may increase runway capacity.

Integrated models: There are several models that solve problems involving individual components of airport operations relatively effectively. However, a major challenge is to form an integrated model. Possible types of integration include integrating runway scheduling, ground movement control, and gate assignment. Another example is the scheduling of an aircraft's take-off and landing at the same time, which requires runways at several airports to be scheduled simultaneously.

Throughput is the primary objective for ATC: The literature considers many different objective function criteria, whereas controllers are generally only concerned with throughput, after safety considerations are taken into account. In order to balance other criteria, controllers need more information and good decision support tools to use this information.

Availability of information in advance: The accuracy and timeliness of information can improve decision making. An aim of the Collaborative Decision Making (CDM) approach for airports is to provide relevant information to all parties (airport, airlines, and ATC) in advance. This helps controllers to schedule landings and take-offs with better insight into the future state of the system.

US versus Europe: There is greater research activity in airport runway scheduling in the US as compared to Europe. The number of available runways per airport in the US is generally more than in Europe, which may affect the characteristics of the scheduling model. The difference in the type of research on the ALP and ATP in the US and Europe indicate that joint research projects would provide a good opportunity for both communities to enhance their models and further develop their solution algorithms.

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